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BYD MEGA MOTOR CO.

INSTALLATION OF MMC/BYD EV ASSEMBLY PLANT
AT GHARO

REQUEST FOR QUOTATION (RFQ)
FOR
LV DISTRIBUTION PANELS



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Rev.	Date	Description	Prepared By	Checked By	Approved By

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BILL OF QUANTITIES

BILL OF QUANTITIES**A. Preamble:**

1. The Bill of Quantities shall be read in conjunction with the Conditions of Contract, Specifications and Drawings.
2. The quantities given in the Bill of Quantities are estimated and provisional, and are given to provide a common basis for bidding. The basis of payment will be the actual quantities of work executed and measured by the Contractor and verified by the Engineer and valued at the rates and prices as given in the priced Bill of Quantities, where applicable, and otherwise at such rates and prices as the Engineer may fix as per the Contract.
3. The rates and prices as given in the priced Bill of Quantities shall, except insofar as it is otherwise provided under the Contract include all costs of Contractor's plant, labour, supervision, materials, execution, insurance, profit, taxes and duties, together with all general risks, liabilities and obligations set out or implied in the Contract. Furthermore, all duties, taxes and other levies payable by the Contractor under the Contract, or for any other cause, as on the date 28 days prior to deadline for submission of Bids, shall be included in the rates and prices and the total Bid Price submitted by the Bidder.
4. A rate or price shall be entered against each item in the priced Bill of Quantities, whether quantities are stated or not. The cost of items against which the Contractor will have failed to enter a rate or price shall be deemed to be covered by other rates and prices entered in the Bill of Quantities.
5. The whole cost of complying with the provisions of the Contract shall be included in the items provided in the priced Bill of Quantities, and where no items are provided, the cost shall be deemed to be distributed among the rates and prices entered for the related items of the Works, or the same shall be determined by the Engineer.
6. General directions and description of work and materials are not necessarily repeated nor summarised in the Bill of Quantities, therefore references to the relevant sections of the Bidding Documents shall be considered. Before entering prices against each item in the Bill of Quantities, the bidders must carefully study in detail, the reference drawings/documents to understand the complete scope of work against each BOQ item(s).
7. Provisional sums included and so designated in the Bill of Quantities shall be expended in whole or in part at the direction and discretion of the Engineer.

BILL OF QUANTITIES

B. Work Items:

The Bill of Quantities contains the following Bills and Schedule

1. 365-01-BQG-001 - General Items
2. 365-01-BQE-009 - Bill Of Quantities for Distribution Step Down Transformer
3. 365-01-BQG-002 - Miscellaneous Items
4. Summary of Bill of Quantities

Bidders shall price the Bill of Quantities in Pak. Rupees (PKR) only.

BILL OF QUANTITIES**365-1-BQG-001- General Items**

Item	Description	Specific Provisions Ref.	Unit	Quantity	Rate		Amount PKR
					PKR In figures	PKR In words	
1	2	3	4	5	6		7
1.	Insurances of the Works	SP	LS				
2.	Insurance of the Goods	SP	LS				
3.	Professional Indemnity Insurance	SP	LS				
4.	Third party insurance	SP	LS				
5.	Workmen Compensation Insurance	SP	LS				
6.	Performance Security	SP	LS				
7.	Price Adjustment	SP	PS				
8.	Provide and equip Engineer's offices	SP	LS				
9.	Maintain Engineer's offices	SP	LS				
10.	Provide diversion road	SP	LS				
11.	Provide for cleaning up the Site on completion	SP	LS				
12.	*Provide for other facilities for Engineer and Employer's Personnel		LS				
13.	Contingencies		PS				
14.	DAAB Fee		PS				
Total for 365-1-BQG-001_____ (Carried forward to Summary Page)							

BILL OF QUANTITIES FOR
LV DISTRIBUTION PANELS

MEGA MOTOR COMPANY PVT. LTD.
INSTALLATION OF MMC / BYD EV ASSEMBLY PLANT AT GHARO
ELECTRICAL BOQ FOR LV DISTRIBUTION PANELS

SR#	DESCRIPTION	UNIT	QTY	SUPPLY UNIT RATE (PKR)	SUPPLY TOTAL RATE (PKR)	INSTALLATION UNIT RATE (PKR)	INSTALLATION TOTAL RATE (PKR)	TOTAL AMOUNT (PKR)	REMARKS / REFERENCE
		A	B	C	D=BXC	E	F=BXE	G=D+F	
1.0	LOW VOLTAGE POWER DISTRIBUTION PANELS								
1.1	SUPPLY, INSTALLATION, TESTING & COMMISSIONING OF FOLLOWING FLOOR MOUNTED 0.4 KV, 50 HZ LOW VOLTAGE POWER DISTRIBUTION PANEL TYPE TESTED, FORM 4B TYPE 6, 12 SWG, FULL RATED TPN+E BUSBAR, IEC-60947-2 COMPLIED BREAKER WITH COMPLETE GADGET AND METERING AS PER REFERRED SCHEMATICS AND SPECIFICATIONS AND SHALL BE COMPLIANCE WITH IEC 61439, BS 61439, IEC 60529 AND RELEVANT IEC & GB STANDARDS INCLUDING FABRICATION, FIXING / BOLTING OF GALVANIZED STEEL BASE FRAMES FOR PANEL MOUNTING WITH ALL ASSOCIATED MOUNTING AND INSTALLATION ACCESSORIES INCLUDING DELIVERY AT SITE WITH LOADING AND UNLOADING THROUGH CERTIFIED CRANE WITH OPERATOR AND RIGGERS. COMPLETE IN ALL RESPECTS. CONTRACTOR SHOULD SUBMIT THE SHOP DRAWINGS FOR APPROVAL BEFORE COMMENCEMENT OF WORK. FURTHERMORE, SCOPE ALSO INCLUDES ALL EARTHING CONNECTIONS OF THE PANELS / AUXILIARIES COMPLETE WITH EQUIPOTENTIAL BONDING OF THE SYSTEM COMPLETE IN ALL RESPECTS								SCHEMATICS 1409A5DQ1A-36G01_1T1 1409A5DQ1A-38G01_1T3 1409A5DQ1A-39G01_1T4 1409A5DQ3A-28G03_3T1 1409A5DQ3A-29G03_3T2
1.1.1	1T1-1AN1	1	NO.						BODY WELDING SHOP (1T1)
1.1.2	1T1-1AN2	1	NO.						
1.1.3	1T1-1AN3	1	NO.						
1.1.4	1T1-1AN4	1	NO.						
1.1.5	1T3-3AN1	1	NO.						
1.1.6	1T3-3AN2	1	NO.						BODY WELDING SHOP (1T3)
1.1.7	1T3-3AN3	1	NO.						
1.1.8	1T3-3AN4	1	NO.						
1.1.9	1T3-3AN5	1	NO.						
1.1.10	1T3-3ALL1	1	NO.						
1.1.11	1T4-4AN1	1	NO.						BODY WELDING SHOP (1T4)
1.1.12	1T4-4AN2	1	NO.						
1.1.13	1T4-4AN3	1	NO.						
1.1.14	1T4-4AN4	1	NO.						
1.1.15	1T4-4ALL1	1	NO.						
1.1.16	3T1-1AN1	1	NO.						G.A SHOP (3T1)
1.1.17	3T1-1AN2	1	NO.						
1.1.18	3T1-1AN3	1	NO.						
1.1.19	3T1-1AN4	1	NO.						
1.1.20	3T1-1AN5	1	NO.						
1.1.21	3T2-2AN1	1	NO.						G.A SHOP (3T2)
1.1.22	3T2-2AN2	1	NO.						
1.1.23	3T2-2AN3	1	NO.						
1.1.24	3T2-2AN4	1	NO.						
2.0	POWER FACTOR IMOPROVEMENT PANELS								
2.1	SUPPLY, INSTALLATION, TESTING & COMMISSIONING OF FOLLOWING 0.4KV, 50HZ AUTOMATIC/MANUAL POWER FACTOR IMPROVEMENT PANELS (PFI) CONTRACTOR IS RESPONSIBLE FOR LOW VOLTAGE DISTRIBUTION PANELS STEPS CONFIGURATION SHALL BE (12.5,25,50,100) KVAR WITH LOAD BREAK SWITCH. INCLUDING POWER FACTOR CONTROLLER AND METER. THIS SHALL INCLUDE ALL INSTALLATIONS, MOUNTING AND FIXING ACCESSORIES. THE PFI PANELS CUBICLES HEIGHT SHALL BE SAME AS THE LV PANEL AS PER REFERRED SCHEMATICS INCLUDING DELIVERY AT SITE WITH LOADING AND UNLOADING THROUGH CERTIFIED CRANE WITH OPERATOR AND RIGGERS. COMPLETE IN ALL RESPECTS. CONTRACTOR SHOULD SUBMIT THE SHOP DRAWINGS FOR APPROVAL BEFORE COMMENCEMENT OF WORK. FURTHERMORE, SCOPE ALSO INCLUDES ALL EARTHING CONNECTIONS OF THE PANELS / AUXILIARIES COMPLETE WITH EQUIPOTENTIAL BONDING OF THE SYSTEM COMPLETE IN ALL RESPECTS								SCHEMATICS 1409A5DQ1A-36G01_1T1 1409A5DQ1A-38G01_1T3 1409A5DQ1A-39G01_1T4 1409A5DQ3A-28G03_3T1 1409A5DQ3A-29G03_3T2
2.1.1	1T1-1ACC1 & 1T1-1ACC2 (405 KVAR)	2	NOS.						BODY WELDING SHOP (1T1)
2.1.2	1T3-3ACC1 & 1T3-3ACC2 (315 KVAR)	2	NOS.						BODY WELDING SHOP (1T3)
2.1.3	1T4-4ACC1 (405 KVAR)	1	NOS.						BODY WELDING SHOP (1T4)
2.1.4	3T1-1ACC1, 3T1-1ACC2, 3T1-ACC3 (315 KVAR)	3	NOS.						G.A SHOP (3T1)
2.1.5	3T2-2ACC1 (180 KVAR)	1	NOS.						G.A SHOP (3T2)
TOTAL PRICE (PKR)									
NOTES: 1. CONTRACTOR SHALL PROVIDE TECHNICAL LITERATURE, INSTALLATION AND MAINTENANCE MANUALS FOR THE OFFERED EQUIPMENT. 2. BOQ SHALL BE READ IN CONJUNCTION WITH DRAWINGS AND SPECIFICATIONS. 3. CONTRACTOR SHALL BE RESPONSIBLE FOR THE FAT OF THE SUPPLIED EQUIPMENT AT MANUFACTURER'S FACILITY INCLUDING ALL FARES WITH BOARDING AND LODGING. 4. CONTRACTOR TO SEPARATELY MANAGE & MAINTAIN BULK QUANTITY ITEMS RELATED TO ELECTRICAL TO FACILITATE THE MATERIAL RECONCILIATION TOWARDS THE END OF THE PROJECT. 5. CONTRACTOR SHALL VERIFY ACTUAL QUANTITY OF ALL ITEMS AS PER SITE REQUIREMENT. 6. CONTRACTOR SHALL SPECIFY THE COUNTRY OF ORIGIN OF THE QUOTED ITEMS. 7. ANY ITEM/EQUIPMENT FOUND DEFECTIVE AT SITE SHALL BE REPLACED BY THE CONTRACTOR WITHOUT ANY COST. 8. CONTRACTOR SHALL REQUIRE TO QUOTE RESPECTIVE SERIAL SEPARATELY AND COMPLETELY. 9. CONTRACTOR SHALL SPECIFY THE DELIVERY PERIOD IN ITS PROPOSAL.									

BILL OF QUANTITIES

365-1-BQG-002-Miscellaneous Items

Item	Description	Technical Provisions Ref.	Unit	Quantity	Rate		Amount PKR
					PKR In figures	PKR In words	
1	2	3	4	5	6		7
Total for 365-1-BQG-002_____ (Carried forward to Summary Page)							

**BILL OF QUANTITIES
SUMMARY**

Bill No.	Bill	Amount(PKR)
1.	General Items (365-01-BQG-001)	
2.	Bill Of Quantities for LV Distribution Panels (365-01-BQE-009)	
3.	Miscellaneous Items (365-1-BQG-002)	
Sub-Total of BOQs		
Federal/Provincial Sales Tax on Services (%) [To be entered by the Employer]		
Grand Total		

Note: All Provisional Sums are to be expended in whole or, in part at the direction and discretion of the Engineer.

GENERAL TECHNICAL

REQUIREMENTS

CONSTRUCTION SCHEDULE

The whole of the Works, and each Section (if any), shall be completed within the Time for Completion for the Works or Section (as the case may be) stated as hereunder and mentioned in Contract Data:

Description	Time for Completion(days)
a) Whole Works	
b) Section A	
c) Section B	
d) _____	
e) _____	

[The Bidder shall provide, the Construction Schedule in the bar chart (CPM, PERT or any other to be specified herein) showing the sequence of work items and the period of time during which he proposes to complete each work item in such a manner that his proposed programme for completion of the whole of the Works and Sections of the Works may meet Employer's completion targets in days noted above and counted from the Commencement Date (Attach sheets as required for the specified form of Construction Schedule)]

METHOD OF PERFORMING THE WORK

[The Bidder is required to submit a narrative outlining the method of performing the Work. The narrative should indicate in detail and include but not be limited to:

1. Organization Chart:

Shall indicate head office and field office personnel involved in management and supervision, engineering, equipment maintenance and purchasing.

2. Mobilization:

In Pakistan, the type of facilities including personnel accommodation, office accommodation, provision for maintenance and for storage, communications, security and other services to be used.

3. Method of executing the Works:

The procedures for installation of equipment and machinery and transportation of equipment and materials to the site.]

LIST OF MAJOR EQUIPMENTS

[The Bidder will provide a list of all major equipment and related items, under separate heading for items owned, to be purchased or to be arranged on lease by him to carry out the Works. The information shall include make, type, capacity, and anticipated period of utilization for all equipment.

The Bidder shall provide adequate information to demonstrate clearly that it has the capability to meet the requirements for the key equipment listed.

LIST OF MAJOR EQUIPMENT
Owned, Purchased or Leased

Owned Purchased or Leased	Description of Unit(Make, Model, Year)	Capacity HP Rating	Condition	Present Location or Source	Date of Delivery at Site	Period of Work on Project
1	2	3	4	5	6	7
a. Owned						
b. To be Purchased						
c. To be arranged on Lease						

LIST OF MAJOR EQUIPMENT**Equipment details**

Item of equipment		
Equipment information	Name of manufacturer	Model and power rating
	Capacity	Year of manufacture
Current status	Current location	
	Details of current commitments	
Source	Indicate source of the equipment ☐ Owned ☐ Rented ☐ Leased ☐ Specially manufactured	
The following information shall not be applicable for equipment owned by the Bidder		
Owner	Name of owner	
	Address of owner	
	Telephone	Contact name and title
	Fax	Telex
Agreements	Details of rental / lease / manufacture agreements specific to the project	

[This Table shall be used for each item of Equipment separately]

**ORGANIZATION
CHART FOR THE
SUPERVISORY STAFF AND LABOUR**

LIST OF SUB-CONTRACTORS

I/ We intend to subcontract the following parts of the Work to sub-contractors. In my/ our opinion, the subcontractors named hereunder are reliable and competent to perform the part of the work for which each is listed.

Enclosed are documentation outlining experience of subcontractors, the curriculum vitae and experience of their key personnel who will be assigned to the Contract, equipment to be supplied by them, size, location and type of contracts carried out in the past.

Part of Works (Give Details)	Subcontractor (With Complete Address)
1	2

ESTIMATED PROGRESS PAYMENTS

Bidder's estimate of the value of work which would be executed by him during each of the periods stated below, based on his Programme of the Works and the Rates in the Bill of Quantities, expressed in thousands of PKR:

Quarter	Amounts (1,000PKR)
1	2
1 st	
2 nd	
3 rd	
4 th	
5 th	
6 th	
7 th	
8 th	
9 th	
Bid Price	

CONSTRUCTION CAMP AND HOUSING FACILITIES

[The Bidder shall provide description of his construction camp's facilities and staff housing requirements. The Bidder shall list or explain his plans for providing these facilities for the service of the Contract as follows:

- 1. Site Preparation (clearing, land preparation, etc.).*
- 2. Provision of Services.*
 - a) Electrical power (expected power load, etc.).*
 - b) Water (required amount and system proposed).*
 - c) Sanitation (sewage disposal system, etc.)*
- 3. Construction of Facilities*
 - a) Contractor's Office. Workshop and Work Areas (areas required and proposed layout, type of construction of buildings, etc.).*
 - b) Warehouses and Storage Areas (area required, type of construction and layout).*
 - c) Housing and Staff Facilities (Plans for housing for proposed staff, layout, type of construction, etc.).*
- 4. Construction Equipment Assembly and Preparation (detailed plans for carrying out this activity).*
- 5. Other Items Proposed (Security services, etc.)]*

Note:

The Contractor shall be responsible for pumps, electrical power, water and electrical distribution systems, and sewerage system including all fittings, pipes and other items necessary for servicing the Contractor's construction camp.

GENERAL SPECIFICATIONS

1.01 Scope

The scope of work shall include the provision of all materials, labor, equipment, and associated services necessary for the complete supply, installation, testing, and commissioning of the LV Distribution Panels:

The Contractor shall be responsible for providing within his bid price any additional equipment and or make modifications equipment/material to suit the requirement of the equipment offered by him for ensuring proper operation of the system as approved by the Consultant.

The Contractor shall ensure all necessary co-ordination with the works of other Contractor and shall be responsible for any loss or damage caused due to his fault or negligence, and shall rectify the same at his own cost.

- i. Examine the work of other Divisions upon which the work of this Division interfaces with and/or depends for proper completion. Report any defect or variance to the Engineer. Do not commence work under this Division until such defects have been corrected
- ii. Co-ordinate the work of the Electrical Division with the work of the other Divisions in all such a manner to ensure proper co-ordination and that there is no interference. In areas where conduits and equipment called for in the Electrical Division are to be installed in conjunction with pipes, ductwork and equipment called for in other Divisions co-ordinate the work with the other Divisions to ensure the best use of the space.
- iii. Verify in the field dimensions, location and clearances affecting the work of this Division.

All allied and small works and materials, even if not specifically mentioned in the specifications and BOQ but required for completeness of the job, shall be deemed to have been included in the contract/BOQ.

1.02 General

- a) The Contractor shall take care not to damage the existing structures, services and equipment during execution of work. If so done, he shall repair and make good all losses at his own cost.
- b) The Contractor shall take extreme care in maintaining proper supply where required to all working areas during the installation period, so there is no disruption in the Employer's working schedule. All the above shall be done at no extra cost.
- c) It shall be presumed that the spaces and access shown in the tender drawing are adequate and proper for the equipment to be supplied by the Contractor, unless so notified by the Contractor at the time of submission of the bid.

- d) Electrification work shall be carried out by a licensed Electrical Contractor, in conformity with the 17th Edition of the UK IEE “Regulations for Electrical Installations”, the “Electricity Act, 1910”, the “Electricity Rules, 1937”, in accordance with the requirements of the local Electric Inspector, the K.E, the Pakistan
- e) It shall be the responsibility of the Contractor to serve notices on, submit documents to, and to have the installation passed by the relevant Authorities, and obtain all necessary approvals and sanctions at no extra cost to the Employer. Official fees for electrical inspections and approvals shall be paid by the Owner at actual.

1.03 Materials and Workmanship

All materials, equipment & accessories, shall be new. Detailed shop drawings, wherever required, (including dimensional plans, elevations, sections, etc.) shall be approved by the Employer/Consultant before pre-casting/pre-fabrication.

All workmanship shall be first class, and undertaken by workmen, skilled in the particular type of craft. All works shall be supervised by experienced engineers/supervisors.

1.04 Shop Drawings/As-Built Drawings/ Record Drawing

The Contract Drawings are schematic and are intended to enable the Contractor to prepare his estimate and submit a tender.

Checking of the Contractor’s Shop Drawings by the Engineer, shall not relieve the Contractor of his responsibility for compliance with the design intent, co-ordination or with any other obligations and requirements under the Contractor.

The Contractor shall allow in his Tender for providing:

Shop Drawing:

2–Copies of shop drawing for preliminary checking by
Consultant 4– Copies of finally approved shop drawing
for onward transmission

As-Built Drawing

2– Copies of As-built drawing for preliminary checking by Consultant
4–Copies of finally approved As-built drawing + reproducible drawing for
onward transmission

All drawings prepared by suppliers, manufacturers and/or Contractors shall be endorsed with the Contractor’s own name title block and drawing number and the Contractor shall take full responsibility for such drawings in accordance with his obligations under the contract. Sample & detailed manufacturer’s shop drawings (including dimensional plans, elevations, section, foundation details etc.) shall be got approved by the Employer / Consultant. Before pre-casting / pre-fabrication.

Record Drawing:

- Maintain one complete set of record contract white prints on site at all times. As the work progresses, clearly mark in red all site changes and deviations from the Engineer's "Construction Issue" documents. These shall include, but not be limited to, the locations of panels, junction boxes, equipment, underground services, feeders to lighting, distribution, communications, and signal panels, as well as all conduit runs for every system. Record drawings shall also document all existing systems and/or installations to be retained. The level of detail shown shall be equivalent to that required by this Specification had the installation been carried out under this Division.
- In addition to the record contract set, maintain one current and up-to-date set of white prints on site at all times for day-to-day reference.
- Clearly indicate dimensions of all buried services relative to the building's structural grid (column lines), as well as invert levels relative to finished floor levels or site grades.
- Record all revisions and variations resulting from authorized changes, including Change Orders and Engineer's Instructions.
- The record drawings shall be reviewed monthly by the Site Engineer and will be considered during the evaluation of the Contractor's monthly Progress Payment Application.
- All record drawings shall accurately incorporate every approved Change Order and Engineer's Instruction issued during the course of the project.
- All record drawings shall be prepared using AutoCAD 2007 or a later version.
- Submit record drawings in the form of one reproducible copy and three prints for initial review. After review and comment, provide three additional revised prints for record. All drawings shall include adequate blank space on the front page for Engineer's and Contractor's "review" stamps.
- Upon final approval of the record drawings, submit the following to the Engineer:
 - i. Three complete sets of prints
 - ii. One reproducible copy
 - iii. One digital set of files on disks, including complete AutoCAD drawing files, the directory structure used, and all required reference files

1.05 Equipment Protection

Unless otherwise noted, all equipment supplied shall conform, as a minimum, to the following protection classes, in accordance with BS 5490:

Indoor	IP 42
Outdoor	IP 65 / IP 66

With the exception of material specified to be hot-dip galvanized (after fabrication), or otherwise specified, all metal work (steel conduit and accessories, outlet and pull boxes, trunking, straps, brackets, hangers, frames, etc.) shall be given a protective treatment consisting of degreasing, derusting, two coats of zinc-chromate/ red-lead primer, and resin- bonded powder coat finish (70µm thickness, 150µm on edge) during manufacture. After installation on site, any damaged metal work shall be given another coat of paint. Painting of steel conduits shall be to BS 1710 for color coding purposes.

All metal-work located outside buildings shall be protected against corrosion by a hot-dip galvanized coating, to BS 729, before installation.

1.06 Applicable Standards

The standards, codes of practice and recommendations of the following Societies or institutions have either been used in the specification and or cited here as a general level of quality for equipment, material and workmanship.

International Electrotechnical Commission	IEC
National Standard of China	GB
British Standards Institution	BSI
European Committee for Standardization	CEN
International Standards Organization	ISO
Underwriters Laboratories	UL
Institution of Electrical Engineers	IEE
Institute of Electrical and Electronics Engineers	IEEE
Chartered Institution of Building Services Engineers	CIBSE
Electronic Industries Alliance	EIA
National Fire Protection Association	NFPA
American National Standards Institute	ANSI
Building Industry Consulting Service International	BICSI
National Electrical Code	NEC
Pakistan Standards Institution	PSI
Building Energy Code of Pakistan	BECP
Karachi Electric	K.E

1.07 Testing & Commissioning

Stage wise testing of the installation shall be carried out at site before/after commissioning of equipment and shall include the following as a minimum:

- Polarity test for switches, MCB's, MCCB's and fuses
- Earthing resistance tests, and earth loop resistance tests
- Megger tests, on switchgear and cables
- Tests as specified for special electrical systems.
- The control systems shall be tested for correct operation by trial run and simulation of all operating and fault conditions.
- All routine tests on equipment shall be performed at the manufacturer's works in the presence of the Consultant or his representative prior to delivery of equipment. Test results and certificate shall be provided in triplicate, to the Consultants/Owner.
- Operation, testing and commissioning of the entire installation.
- Copies of final test reports and values shall be provided to the Consultant/Employer in triplicate.

1.08 Extra Work

- a) Written approval from the Consultant shall be obtained before any extra / additional work is carried out.
- b) Rate analysis of extra item is to be prepared and enclosed, (on the basis of cost plus 20% overheads, profit, etc.) with the bills.

1.09 Safety Program

- a) The Contractor shall strictly follow at his own cost the Safety Programme outlined below and such additional measures as the Engineer or Engineer's representative may determine to be reasonably necessary.
 - i) Prior to commencement of work the successful Contractor shall submit safety programme for discussion with the Employer and the Engineer.
 - ii) The Contractor shall prepare a plan of the Works' Site to assure that storage areas for materials and equipment are located on the project/worksite for maximum efficiency. This plan will be subject to the approval of the Engineer.
 - iii) Activities between different operations and different crafts will be coordinated.
 - iv) The Contractor shall lay out and provide an efficient access system with information and directional signs posted as necessary.
 - v) All employees will be instructed on safe work method.

- vi) The Contractor shall advise all his supervisory staff of their responsibility for the prevention of injury to persons or damage to property or equipment in the irrespctive areas of supervision.
 - vii) Safety will be included in all job planning. This will include providing safe construction equipment and vehicles, protective equipment necessary for protection of workmen, and establishing methods for safe operation.
 - viii) Good housekeeping will be maintained at all times.
 - ix) Scaffolds, ladders, ramp”, runways etc. will be constructed properly and maintained in safe conditions.
 - x) Ample fire protection will be provided and fire hazards guarded, by the Contractor.
 - xi) Adequate lighting, ventilation etc. will be provided as necessary, by the Contractor.
 - xii) Equipment will be properly and regularly inspected and maintained by the Contractor to the satisfaction of the Engineer.
 - xiii) The Contractor will assign to his employees only such duties as are within their physical and mental capabilities.
 - xiv) The Contractor shall hold monthly meetings with supervisory staff, while the designated person in charge at the lower level shall conduct weekly safety meetings of 10 to 15 minutes with their crew.
 - xv) First Aid facilities will be provided at job sites, the services of doctors and hospitals made available, and all supervisors instructed in handling of injured employees.
 - xvi) Adequate toilet facilities will be provided, maintained in a hygienic condition and their use enforced by the Contractor.
- b) Accident reports will be furnished to the Engineer for onward transmission to the Employer within 2 days of the reported accident.

Copies of the safety programme will be supplied by the Contractor to the Engineer and will be promptly posted in all offices in use of projects/ works under this Contract.

- c) At all construction sites the following instructions shall be followed:
- ◆ Availability of safety (hard) hats, which should be worn by anyone visiting or working within the designated work area.
 - ◆ controlled entry to the designated work area.
 - ◆ Proper distribution of temporary electric power (use of RCDs /E LCBs, switchgear, cabling, socket outlets)
 - ◆ Proper guarding of shafts, stair sand floor edges up, to a height of 42”.
 - ◆ Proper access stairs and ladders with handrails.

- ◆ Properly demarcated passage ways, which are kept clear of materials, equipment, and rubbish.
- ◆ Daily cleaning of debris and rubbish from the site.
- ◆ Adequate temporary lighting.
- ◆ Proper house keeping to reduce slipping and tripping hazards.
- ◆ Proper protection to pedestrians and adjoining buildings.
- ◆ Contractor's All-risk Insurance policy to be in place.

SPECIFICATIONS FOR LOW VOLTAGE SWITCHGEAR

(365-01-SPE-006)

1.0 **INTRODUCTION**

1.1 **Scope**

This specification covers the minimum basic requirements for the engineering, design, manufacturing, inspection, testing, commissioning and supply of 400V Low Voltage Switchgears to be installed indoor.

The scope of this Specification includes specific technical requirements for indoor switchgears & MCCs, such that, the design, manufacture, testing and performance of equipments, and components, included in this specification shall comply with all applicable IEC Standards / Documents (latest issues, amendments, and supplements to these documents shall apply, unless otherwise indicated), as listed below:

1.2 **Definitions**

Following definitions apply throughout this document:

Project Owner / Client	: BYD/Mega Motor Co..
Project Name	: Installation of BYD/MMC EV Assembly Plant at Gharo
Engineering Consultant	: Zishan Engineers (Pvt.) Ltd. Karachi, Pakistan

1.3 **Errors or Omissions**

Any errors or omissions noted by the Supplier in this Specification shall be immediately brought to the attention of Company.

1.4 **Deviations**

All deviations to this Specification, other related specifications or attachments shall be brought to the knowledge of the Company in the bid. Such deviations shall be shown in the documentation submitted to Company/Consultant.

1.5 **Conflicting Requirements**

In the event of any conflict, inconsistency or ambiguity between the Contract scope of work, this Specification, National Codes & Standards referenced in this Specification or any other documents, the Supplier shall refer to the COMPANY / CONSULTANT whose decision shall prevail.

2.0 **CODES AND STANDARDS**

- IEC 60947-2 Low Voltage Switchgear and Control gear, Part: 2 - Circuit Breakers
- IEC 61439-1 Low Voltage Switchgear and Control gear Assemblies, Part: 1 – Type Tested and partially Type Tested Assemblies
- IEC 60529 Degree of Protection provided by Enclosures (IP Code)
- IEC 60947-4 Contactor and Motor starter
- IEC 60947-3, part 3 (Switch disconnectors and fuse combinations)
- IEC 60747-7 Auxiliary equipment-Terminal blocks
- IEC 61439-6 Busways
- IEC 61439-3 Distribution Board
- IEC 61800-3 Adjustable speed electric power device
- IEC 60715 Dimension of LV switchgear and Control gears
- IEC 60258 Electrical measuring units and accessories
- IEC 60255-1 Measuring relay and protection equipment
- IEC 60151 Over/Under Current relay
- IEC 60149 Thermal relay
- IEC 60127 Over/Under Voltage relay
- IEC 60181 Frequency relay (Hz)

3.0 **GENERAL**

Switchgears & MCCs shall comply with, and configured and wired as detailed in IEC 61439-1.

Measurements shall be in SI Units.

Documents shall be in English.

Rated Voltage and Current for Switchgears & MCCs shall be based on 50 Hz.

4.0 **GENERAL ELECTRICAL CHARACTERISTICS**

The Switchgears shall be as per the following Electric Supply System:

- Voltage : 400 Volts, $\pm 5\%$
- Frequency : 50 Hz, $\pm 2\%$
- Phase / Wire : 3 Phase / 4 Wire
- Fault Level : As per calculation
- Neutral Grounding : Solid

The Switchgears & MCCs shall be equipped with ACBs (Draw out Type), MCCBs (Fixed Type), Motor / MCCBs / Contactors (Draw out Type), as indicated on respective SLDs and shall be type tested.

Each incoming breaker and BUS Coupler shall be furnished with dedicated electronic protection relay modules (EPRs) covering the minimum protection features & LSIG units as shown in relevant single line diagram.

All EPRs shall be connected in redundant loop.

The Cable Entry shall be of “Bottom” Type.

The Bus Duct Entry shall be of “Top” Type if any.

The Main Bus Bars, Risers and Earth Bus shall be of “Electrolytic grade Copper” Material.

The Painting shall be of “EPOXY” Type.

The Space Heater, and Auxiliary / Control supply voltage for all of the outgoing circuits shall be 230 V AC. However for Incoming ACB's and Bus Coupler Control Supply shall be TBC VDC / VAC.

The Enclosures Degree of Protection shall be at least IP42, as defined in IEC 60529.

Feeders equipped with Power Transducers (as per SLDs) shall be of “Class 0.5”.

All Control Bus Bars shall be of “Tinned Copper”.

All ACBs shall be equipped with the following Features / Characteristics:

- DSF
- Sealing Cover
- Shunt Release, TBC VDC / V AC
- Safety Shutter & Operation Counter with Closing Coil, TBCVDC / V AC
- Service Position Switches
- 230 V AC Spring Charged Motors
- Cut Off Switches
- Spring Charged Close Signal Switch
- Auxiliary Contact Switches
- Door Interlocking Kits
- Under Voltage Release (UVR) 230V AC

All Motor Feeders shall be equipped with separate Contactor for Instrument Interlock, and Instrument Ready Indication.

Withdraw able type starter / modules of same rating shall be suitable for interchange ability in between panel enclosures

Current Transformers and Potential Transformers shall be of Cast Resin Type only.

All Motor Feeders shall be equipped with Test Lamp, Push Button, and Inst. Ready Indication Lamp.

The Control Transformer (400V / 230V) shall be Electrically Interlocked, through Contactors of suitable size.

All Motor Feeder Ammeters & Voltmeters shall be digital type.

Switchgears & MCCs shall be complete with all necessary cable terminations, cable spreader boxes etc., as required to complete the works.

Push Buttons shall be of heavy-duty type. Indication Lights / Lamps shall be push to test type, with wide band LED Pilot Light only with built-in current limiting resistor & reverse polarity protection diode.

Control Relays contacts shall be rated for minimum 10 amp and with 230 V AC operating coils. The number and type of contacts shall be as required to perform the operations specified.

All ACBs shall have service, test and ground positions. ACBs shall have provision of manual charge in addition to motorized charging.

All motor feeders shall have reset button on front plate of feeder to reset thermal relay using Bowden cable.

Selector switches shall be of rotary type with enclosed contacts and shall have adequate number of contacts to perform their switching functions properly.

All units designated “Spare” on the Single Line Diagrams”, shall be furnished completely wired, and fully equipped with the breakers for the sizes, as designated on SLDs.

Power circuit components shall be rated for the specified operating voltage. The equipments shall be capable of proper operation for voltage deviations of plus minus 10%, and frequency deviations of plus minus 5%.

5.0 **OTHER CHARACTERISTICS (Common Enclosure and Cubicle Design)**

The Switchgears & MCCs should be indoor type, free standing, bottom cable entry, cubicle type modular construction, and front operated.

The Switchgears & MCCs shall be designed, and constructed to “Form 4B” 6, type tested, as defined in IEC 61439 Standard, such that the failure of one equipment does not affect the adjacent units. Between bus compartment and breaker compartment, and breaker compartment and cable compartment, non- perforated flame retardant partitions shall be provided.

The switchgears & MCCs shall be type tested for mechanical resistance test, which ensures the strength of the structure against electro dynamic forces due to short circuit currents.

The switchgears & MCCs shall be type tested for internal arc withstand as per IEC 61641.

All live parts shall be isolated from direct contact by barriers, and enclosures.

Covers with access to live parts shall be marked as “Danger – Live Parts”. All components shall be labelled with engraved laminated labels.

Properly sized and labelled terminal blocks shall be provided for all connections. All wires and cables shall be terminated with compression type connections, and pins for control wires.

All current carrying parts shall be capable to carry the specified rated current continuously. All de-rating factors shall be allowed for, in the design of the Switchgears & MCCs.

A Copper Earth Bar of appropriate length shall be provided in each Switchgear & MCC.

Relays, meters and control switches shall be located at a height, which shall be convenient for monitoring.

The draw-out mechanism shall hold the breaker rigidly in the service, test, or isolated / disconnected positions. It shall be possible, when the breaker is open to position the breaker in any of the three positions with the door closed.

The circuit breakers and enclosures shall be so interlocked as to prevent removal or re-insertion of the breaker without being in the open position, but the door of each cubicle shall be arranged to permit door opening with the breaker closed.

Each breaker shall have a locking device to lock the breaker in the open position.

Stored-energy closing springs shall be automatically discharged before the breaker element may be withdrawn from the cubicle.

Busbar alley shall be covered with screwed perforated sheet to avoid direct access to the vertical busbars on opening of the busbar alley door.

Power and Control terminals in cable alleys (min. 350mm wide), for each module shall be covered with bolted type, sloped hylam sheet or FRP sheet which shall be fixed after termination of power cable.

The Switchgears & MCCs enclosures shall have a metal thickness of approx. 2mm, and additional stiffeners.

The Switchgears & MCCs shall be dead front, and dead back construction, with all required access, provided from the front.

The Switchgears & MCCs enclosures shall be minimum IP42 according to IEC 60529, gasketed, and installed in ventilated or air-conditioned electrical equipment rooms located in unclassified areas.

Push buttons, selector switches, and indication lights / lamps shall be installed on approx. 2mm steel device panel.

Separate pan type removable steel doors shall be used for each cubicle. Doors with neoprene rubber gasket shall be hinged to the structure on the left side with concealed hinges, and shall be held with captive screws on the right side.

6.0 **TERMINAL BLOCKS, WIRING AND GROUNDING**

Adequate space shall be provided for wiring. All wires shall be properly arranged in wiring gutters or control trays.

Control wiring shall be carried out with flexible heat resistant wires of minimum size 1.5mm² for control circuits, and 2.5mm² for CT circuits. Wires connected to earth shall be of green color only.

Each wire shall be identified at both ends with wire designation in accordance with the wiring diagram developed from approved control schematics. Interlocking type plastic ferrules of yellow color shall be used for identification. Color of lettering of ferrule shall be black.

Wires forming part of tripping circuit of circuit breaker shall be provided with additional red ferrule, marked "T", or some other acceptable identification mark shall be provided as per manufacturers standard.

All spare contacts of relays, switches and other components shall be wired up to terminal blocks.

Terminal blocks of different voltage classes shall be segregated. Minimum clear space of 300mm shall be provided between gland plate, and terminal block.

The terminal block shall be grouped and segregated according to circuit functions and voltage, and shall have 20% spare terminals. Individual terminals in each block, in each group shall be serially numbered in accordance with the drawings.

Gland plates shall be of non-magnetic material for all single core cables. Gland plate shall be removable bolted/screwed supported by gasket and thickness shall be in accordance with nos. of feeders to facilitate load of power and control cable glands without causing any sag.

Motor Starter Feeders / Units shall be grounded in accordance with IEC 60439-1.

Switchgears & MCCs shall be designed and constructed to ensure proper grounding of the individual units. Maximum resistance between the individual Motor Starter Enclosure and Switchgear ground bus shall not exceed 0.1ohms, as per IEC 61439.

7.0 **BUSES**

Phases & neutral for buses and component terminals shall conform to IEC standards for Switchgears & MCCs.

Horizontal and vertical buses shall be braced to withstand the minimum RMS symmetrical fault current, at a utilization voltage of 400 V.

Bus ratings shall be based on 40°C rise over a 60°C ambient.

Bus bar spacing and the spacing of bus fastening point shall not be less than specified in IEC 61439-1.

The bus bar ratings of the main buses for switchgears & MCCs shall be as per Single Line Diagrams.

Each Switchgear and MCC section shall contain four main horizontal bus bars (L1, L2, L3 & N) housed in an isolated compartment, which shall run continuously through all sections of Switchgears & MCCs.

The main horizontal buses shall be continuous and unspliced.

Each vertical section shall contain four (L1, L2, L3 & N) full length vertical bus bars connected to the horizontal main bus bars at the top of each vertical section.

The vertical buses shall be rated for the connected load.

A continuous, horizontal Copper earth bus shall be located across the complete length of Switchgears & MCCs line up.

A vertical earthing bus shall be provided in each vertical section that makes contact with the plug-in units before the bus stabs engage the vertical bus.

Earth bus shall be rated to carry the maximum fault current.

Hinged doors shall be earthed through flexible earthing braids. All non-current carrying metal parts shall be effectively bonded to the earth bus.

Standard colour coding / marking on buses will be applicable as per relevant codes & standards.

8.0 **MOTOR STARTER UNITS**

8.1 **Requirements**

- Combination motor starter units / distribution feeders shall be withdraw able type.
- Every motor starter feeder shall be provided with:
 - Test lamp P.B. & Instrument ready indication lamp
 - Indicating lamps (clustered LED type): ON, OFF, TRIP
 - Manual reset of thermal overload relay
 - ON, OFF, TRIP potential free contacts (to DCS)
 - Terminals for ON, OFF, ESD from PLC
- All motor starter units / distribution feeders shall be equipped with digital Ammeters, and selector switches, for motors rated up to 75kW; whereas motors rated above 75kW shall be additionally equipped with kWh, and Volt meters.
- All motor feeders shall have reset button on front plate of feeder to reset thermal relay using Bowden cable.
- Withdraw able type starter / modules of same rating shall be suitable for interchangeability in between panel enclosures

8.2 **Main Disconnect Device Mechanism**

- The main disconnect device shall be MCCB. MCCBs shall be suitable for induction motor operation with magnetic trip protection only and overload protection to be covered via thermal overload relay
- MCCBs shall be equipped with electronic trip feature including both magnetic & thermal protection for all power feeder circuits.

8.3 **Contactor**

- The motor starter contactor shall be a heavy duty magnetically operated, rated for the maximum continuous current of the starter, it will be installed in.

- In addition to auxiliary control contacts required for starter operation, it shall be provided with two additional nos. of auxiliary contacts. These additional terminals shall be wired to the starter control terminal box. Main & Aux contactor shall be selected as per utilization category specified by IEC. Refer SLD for type of load.

9.0 **SPACE HEATERS**

- A minimum of one space heater per section shall be provided.
- Heater shall be rated for continuous duty at 230 V.
- When required, an additional normally closed auxiliary contact shall be provided on the motor starter contactor to allow motor space heater operation, when the motor is not in operation.

10.0 **POWER FACTOR IMPROVEMENT PANEL**

The power factor regulator combines comprehensive operation with Manual & automatic switching operation along with Microprocessor/PLC based controlled relay. With following minimum requirement as below. Panels & components shall be type tested. PFI shall be 12 stage relay controls, LED/HMI display, Measurement and display of key parameters: Voltage, Current, Power factor, THDI etc. shall be capable for RS485 Communication MODBUS RTU Protocol, Reconnection time is same as discharge time. Power factor improving panel shall be capable for maintains THD level as per IEC standard. Harmonic and Power Factor correction panel shall be independent or combined. As per relevant SLD and BOQ (365-01-BQE-009).

- Indoor, Floor mounted, cable entry from bottom.(or specified in detail engineering stage)
- Incomer LBS.
- Step protection
- Step switching
- Capacitor duty contactors with Damping Resistors
- Microprocessor controlled relay
- Switching option Auto& Manual option.
- Copper Bus bar

- Protection class IP42
- Copper bus bar
- Harmonics filters
- Capacitors Heavy Duty can type or equivalent.
- Minimum joints hence better reliability and lower losses.
- Or other requirements as per IEC& NEC standard.

11.0 **BUS TIE DUCT**

Type tested busbar shall be high-current in sandwich design. The busbar for high protection in harsh ambient conditions, e.g., for outdoor networking of building sections. Distribute power over long distances horizontally as well as vertically in industrial plants. Tap-off units with communication capable measuring and switching devices, degree of protection shall be IP66 for power transmission available, with tested fire barrier .Rated insulation operational voltage shall be 1000VAC. Connection shall be Hook and bolt Conductor material shall be copper. With high short circuit ratings

12.0 **INSPECTION AND TESTING**

Inspection and Testing shall be carried out at Manufacturer's works, by the Vendor to ensure that the equipments comply with all the specified requirements, and Purchase Order, in particular that the construction, and wiring requirements and enclosures' classifications are met, and that the components (i.e. CTs, VTs, Relays, Instruments, Lamps, and Controls) are complete, and correct in rating and range.

Routine Tests shall be carried out, as per relevant IEC Standards, at Manufacturer's works, on complete Switchgears & MCCs Assemblies. Routine Tests shall include, but not limited to the following:

- i) Mechanical operation of circuit breakers, outgoing feeder modules, auxiliary switches, manual devices etc.
- ii) Dielectric withstand test
- iii) Primary& secondary injection testing
- iv) Protection relay tripping test
- v) Operational and safety interlocks.
- vi) Operation of circuit breaker for minimum allowable control voltage.

- vii) Closing and opening of circuit breaker at rated and minimum control voltage.
- viii) Electrical & Mechanical operation tests & interchangeability of draw out units
- ix) Millivolt drop test across breaker contacts, and between terminals, when housed in cubicle.

Test Certificates, against Type Tests (of relevant switchgear) shall be provided, by the Manufacturer.

12.1 **Witness of Tests**

The Supplier shall submit for Client/ Engineering Consultant's review and approval, a complete plan for the factory acceptance test. The objective of the FAT is to verify that the purchased items meet or exceed the specified requirements. The Supplier shall arrange the Factory Acceptance Test for Client / Engineering Consultant. Expenses including airfare, boarding & lodging transportation shall be borne by the Supplier for the agreed number of Client/ Engineering Consultant's representative

13.0 **IDENTIFICATION AND LABELING**

Each unit and cubicle on Switchgears & MCCs shall have nameplates on the outside face of the door. Replica of the nameplates on outside face of door shall also be placed on the inner side of draw out type module.

The devices within Switchgears & MCCs shall be identified with a nameplate.

The equipment mounted through the doors shall have nameplates provided on both sides of the mounting surface.

The overall Switchgears & MCCs assemblies shall be centrally and visibly marked with a nameplate of letters 18 mm or larger including, Manufacturer's name, model and serial numbers, etc.

The nameplates shall comprise engraved letters in laminated phenolic, with a white outer surface, and black core.

14.0 **DRAWING/DOCUMENTS**

The Manufacturer shall describe in his proposal the characteristics of all components of Motor Starter with schematic drawing. Owner shall have the option to ask for a change in the proposed components.

Two sets of preliminary drawings shall be submitted to the Owner for approval. Each set shall include but not limited to:

- Overall and assembly drawings
- Front face arrangement.
- Power and control diagrams/scheme of each module.
- Technical literature of Thermal relays, MCCB, Contactors and Current Transformers.
- Installation & Operation Manuals
- Panels internal wiring and termination drawings.
- Panel to Panel internal loop wiring

On approval of preliminary drawings the Contractor's Manufacturer shall develop final design drawings incorporating owner's comments, if any, before proceeding with the manufacturing.

Three sets of As-built drawings shall be provided to the Owner in addition to those supplied and shipped with the equipment in the document case.

Comments made by CLIENT/CONSULTANT on drawing submittal shall not relieve VENDOR or SUBVENDORS of any responsibility in meeting the requirements of the specifications. Such comments shall not be construed as permission to deviate from requirements of the Purchase Order unless specific and mutual agreement is reached and confirmed in writing.

In addition to the instructions in standard forms, the MANUFACTURER shall comply with the following additional requirements for Installations, Operating and Maintenance Manuals:

- a. The front cover, spine and inside page shall state the Purchase Order number and MANUFACTURER' S reference number.
- b. The inside front page shall carry an index listing the contents of each section of the manual.
- c. Individual sections shall be complete and shall refer to equipment actually supplied.
- d. Published data shall also be included, including published data for bought in items.
- e. A punch list of "do's" and "don'ts" shall be included.

- f. Full details for installation and setting up shall be included.
- g. Items requiring regular inspection, checking, testing and maintenance shall be listed and the time scale clearly indicated.
- h. Important items shall be cross referenced to other parts of the manual as necessary.

15.0 **ACCESSORIES AND TOOLS**

Following accessories shall be provided with Switchgears & MCCs lineup.

- Manual Spring charging device.
- Breaker rack in / rack out trolley.
- Earth switch / truck for earthing busbars / incomers & outgoing.
- Spanner set incorporating all installed bolts sizes
- Allen key set incorporating all installed head types

16.0 **SHIPPING & PACKING**

Switchgears & MCCs shall be packed securely and anchored to skids. Un-mounted parts shall be suitably boxed, and firmly attached for shipment.

Switchgears & MCCs shall be shipped completely assembled, whenever possible. When this is not possible, shipping splits shall be indicated on the drawings.

Manufacturer/Supplier shall provide such packing of the equipments as is required to prevent their damage or deterioration during transit to their final destination as indicated in the Contract. The packing shall be sufficient to withstand, without limitation, rough handling during transit and exposure to extreme temperatures, salt and precipitation during transit and open storage. The containers for storing equipments should carry a packing list giving details of equipments stored in the container. Packing & Shipment shall confirm the requirement of Project Specification.

17.0 SPARES

Manufacturer/Supplier shall supply spares for two (02) years operation as per requirement of BYD/MMC on quoted price. Manufacturer to share list of critical spares to be maintained at client end. Manufacturer to ensure all supplied components are easily available in market for upcoming 10 years and shall not provide any component going to be obsolete in near future. In the event of termination of production of spares an advance notification should be given to BYD/MMC giving sufficient time enabling it to procure the required spares. In case of a product phase-out spare parts and breakers will be managed and should be available for another 10 years after product discontinuation. And following such termination, furnishing at no cost to BYD/MMC, the blue prints, drawings, special tools and specifications of the spares.

Manufacturer to provide special tools set that are necessary for maintenance and replacement of critical parts of LV panel

One complete set of tools for routine operation, setting and maintenance of LV switchboard including protective relays as recommended by manufacturer of each component shall be furnished within the price quoted for the LV Switchboard. The tools shall be supplied in suitable box.

18.0 GUARANTEE

The Supplier/Manufacturer warrants that the supplied equipment's shall be brand new and manufactured/fabricated as per agreed specifications and free from fault. The Supplier/Manufacturer shall guarantee that the equipment's furnished is free from fault in design workmanship, is of sufficient size and capacity, and of proper material to satisfactorily fulfill the operating conditions specified. Should any defect in design, material, workmanship or operating characteristics develop during the twelve (12) months warranty period, the supplier/manufacturer shall make all necessary alterations, repairs, and replacement of defective equipment, free of charge, and shall pay all transportation involved to and from the installation if the defect or failure to function cannot be corrected, the supplier shall replace promptly free of charge the equipment. The Supplier/Manufacturer further warrants that the supplied equipment's shall be in accordance with industry's recognized codes and standards. The Warranty shall be for a period of twelve (12) months from the date of Commissioning or eighteen (18) months from the date of arrival of equipment's whichever occurs first.

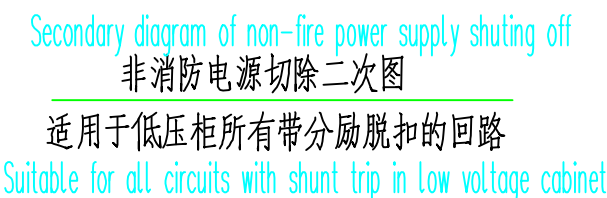
19.0 AFTER SALES & SERVICE

Manufacturer/Supplier has After Sales & service, at least Last 5 years & they have an experience of supplying same & above rating of LV panel.

20.0 **TRAINING**

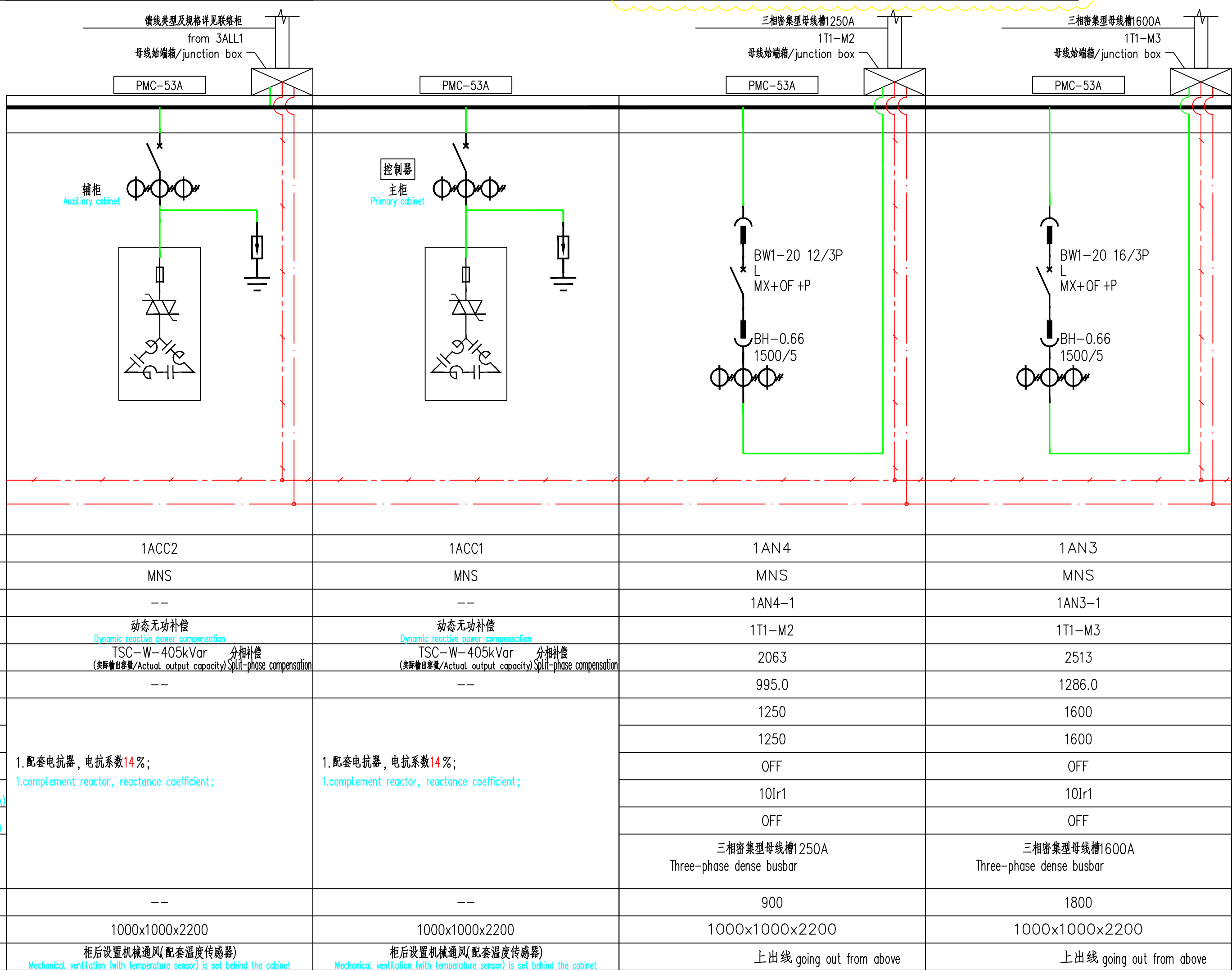
Manufacturer/Supplier conduct onsite and class room training of BYD/MMC operator.
For smooth running of LV panel.

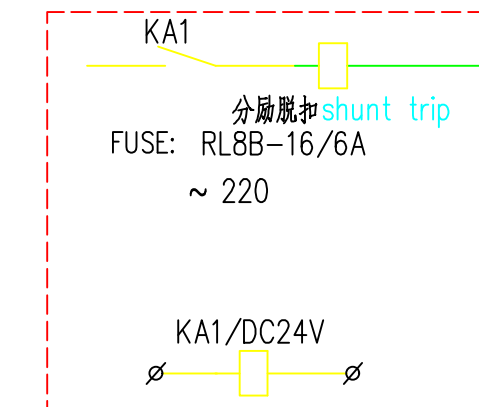
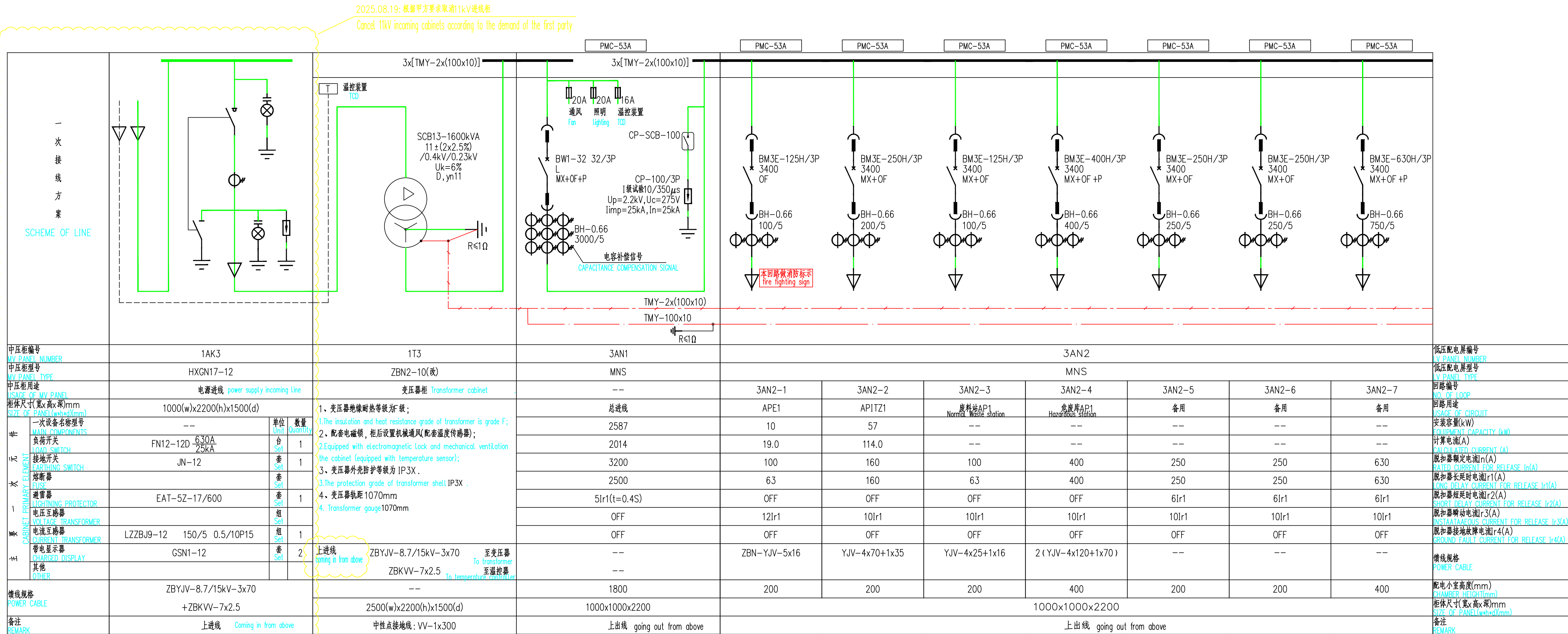
DRAWINGS



1. 干式变压器(带IP3X金属防护外壳)采用电缆进线,零线横槽割出线,横槽出线位置与同相距离应大于槽宽(相配合)。
2. 变压器零线应单独(避免进风),电源接自低压侧开关(加装专用熔断器),温控器零线接警报警线网路电源接ZBKVV-7x2.5mm²。
3. 变压器柜内设置熔断器,电源接自低压侧开关(加装专用熔断器),电压互感器、电容补偿柜后应设置机械式熔断器,电压接自低压侧开关(加装专用熔断器)。
4. 变压器柜门应具有带闭锁功能,变压器零线时应应锁;图中所有断路器均配辅助触点和锁闭装置,锁闭装置必须按业主要求设置,变压器柜门体颜色按业主要求。
5. 变电所地址按生产工厂TN-S制。
6. 柜内二次线路生产采用集成系统柜中的测控表计,按照生产厂家产品样本拟定,本设计不再出图。
7. 低压柜柜门内框架开柜均配进线,进出位置机加安全锁,避免断路器故障时伤人。
8. 低压柜柜门配630A及以上采用抽插式结构,大630A采用固定开闭式结构。
9. 相邻柜间变压器柜应设联锁开关,联络开关与主开关之间设机械联锁与电气联锁,保证三台断路器同时只能有两台合闸。正常时联络开关应断开。
10. 低压柜消防回路两侧设置防火隔板。
11. 电气柜设置温度检测,在柜门显示温度状态,且能接至上级监控系统。

03	2025.08.19	IPC-Capsul 11W incoming cabinets
03	2025.04.03	IPC-Update air conditioning etc.
02	2025.03.03	IFC
01	2024.12.07	FOR APPROVAL
建设单位 CLIENT		
Mego Motor Company (Private) Limited		
项目名称 PROJECT		
Mego Motor Company CXD Factory		
子项名称 BUILDING		
焊装车间		
Body Welding Shop		
图样名称 DRAWING TITLE		
1T1 系统图		
1T1 schematics		
审查 AUTHORIZED		
设计负责人 DESIGN MANAGER		
审核 APPROVAL		
校对 CHECK		
设计负责人 DISCIPLINE LEAD		
设计 DESIGNER		
日期 DATE		
专业: DISCIPLINE	电气	阶段: STAGE
比例: SCALE	1:100	版次: REVISION
图号: DRAWING NO.	1409A5DQ1A-36C01	





Secondary diagram of non-fire power supply shutting off
非消防电源切除二次图
适用于低压柜所有带分励脱扣的回路
Suitable for all circuits with shunt trip in low voltage cabinet

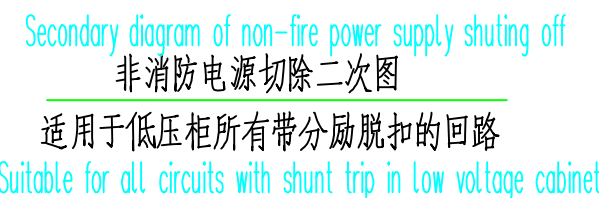
说明:

- 1、干式变压器(带IP3X金属防护外壳)采用电磁锁,接线端子侧出线,接线端子位置与网柜应与网柜屏号对应编制;
- 2、变压器温控器(带远方风冷),电源接自低压侧负序开关(加装专用熔断器),温控器温度信号接网柜号采用KVPW₃-7x2.5mm²;
- 3、变压器屏内设置照明,电源接自低压侧负序开关(加装专用熔断器),变压器柜、电容补偿柜柜内设置机械通风,电源接自低压侧负序开关(加装专用熔断器);
- 4、变压器屏柜上应具有等电位联结功能,所有暴露带电部分应接地;图中所有暴露带电部分应接地端子并确定位置,确定位置来理柜体主要设置位置,配电柜柜体颜色色主要要求;
- 5、变电所接地埋式采用TN-C制。
- 6、柜内二次线路由生产厂厂家提供本系统图中的测控表计,按照生产厂厂家产品样本拟定,本设计不再出图。
- 7、低压配电屏内框架开关和柜体制造/输出位置机械安全锁,要安装防触电警示标志。
- 8、低压配电柜30A及以下采用抽屉式结构,大于30A采用固定分隔式。
- 9、柜内开关柜内设置低压侧联络开关,联络开关与主开关之间设机械联锁与电气联锁,保证三台断路器同时只能有两台合闸,正常时联络开关应断开。
- 10、低压柜前设置回路断路器设置隔板。

Note :

1. Dry-type transformer (with IP3X metal protective shell) use cable lead line, horizontal busbar side outlet, horizontal busbar outlet position and spacing should be matched with the low-voltage cabinet bus.
2. Transformer has a temperature controller (forced air cooling), whose power is from the special fuse added before the main switch of the low-voltage side. The thermostat temperature alarm and trip signal. Lead high-voltage switchgear using $kV\cdot W-72.5mm^2$.
3. Set lighting in the transformer cabinet, whose power is from the special fuse added before the main switch of the low-voltage side. Set mechanical ventilation on the back of the transformer cabinet and capacitor compensation cabinet, whose power is from the special fuse added before the main switch of the low-voltage side.
4. The transformer cabinet door should have a live lock function, and the door should be locked when in live; All circuit breakers are equipped with auxiliary contacts and locking devices, and the type of locking devices is set according to the requirements of the owner. The color of the cabinets is as required by the owner.
5. System of substation grounding type is TN-S.
6. Cabinet secondary circuit is set by the manufacturer combined with the measurement and control meter in the system diagram, and according to the manufacturer's product sample. This design is no longer drawn.
7. In low-voltage cabinet, frame breaker should have remote in/out position mechanical safety lock, molded case circuit breaker should have extension handle.
8. Low voltage distribution cabinets of 630A and below use drawer type structure, of more than 630A use fixed separation type.
9. The two adjacent transformers are equipped with a low-voltage contact switch, and the mechanical interlock and electrical interlock are set between the contact switch and the main switch to ensure that only two of the three circuit breakers can be closed at the same time. The contact switch should be turned off when normal.
10. Set fire-proof partition on both sides of the fire circuit in the low voltage cabinet.

[illegible]

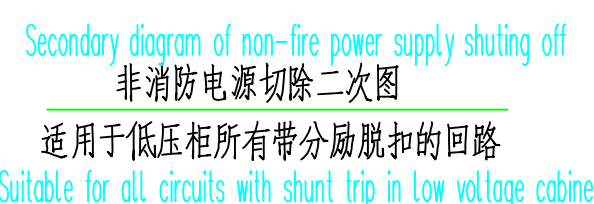


1. 千式互感器(即P3X金属铂壳)采用电缆线及铜导线直接连接, 将输出位置与同组互感器绝缘性能匹配。
2. 变压器零序电压(输入端), 电源接自低压母线(电压: 加表专用电压源), 零序电压源及电压耦合器符合GB/T8947-1997, 220V, 50Hz。
3. 变压器加表专用电压源, 直接接自电压互感器(电压: 加表专用电压源), 变压器、电容补偿柜和电压互感器通风, 电源接自低压侧开关(电压: 加表专用电压源)。
4. 变压器加表专用电压互感器, 变压器零序电压源时间: 所有出厂前需经国家计量检定和定期检定, 额定容量及精度主要要求设置, 额定值在铭牌上标注。
5. 所有受电柜主要设备, 符合下列要求:
 6. 柜中二次回路产生生产厂名称合表图中的控制接线, 按生产厂产品样本拟定, 本设计不再画出。
 7. 低电压电动机断路开关按短路/漏电/漏电故障安全保护, 断路器安全附件选择。
 8. 低电压控制柜: 30A及以下采用插接式结构, 46.350A采用固定式结构。
 9. 柜内所有二次回路电压联锁开关, 联锁开关主分支二次回路设机械联锁(机械联锁, 正常二次断路器时只能有闭合合闸), 正常时联锁开关应处于常开位置。
 10. 低电压保护装置采用熔断器式熔断器。

- 1.Dry-type transformer (with IP3X metal protective shell) use cable lead line, horizontal busbar side outlet, horizontal busbar outlet position and spacing should be matched with the low-voltage cabinet bus.
- 2.Transformor has a temperature controller (forced air cooling), whose power is from the special fuse added before the main switch of the low-voltage side. The thermostat temperature alarm and trip signal lead to high-voltage switchgear using $KVW-7x2.5mm^2$.
- 3.Set lighting in the transformer cabinet, whose power is from the special fuse added before the main switch of the low-voltage side. Set mechanical ventilation on the back of the transformer cabinet and capacitor compensation cabinet, whose power is from the special fuse added before the main switch of the low-voltage side.
- 4.The transformer cabinet door should have a live lock function, and the door should be locked when it is live; All circuit breakers are equipped with auxiliary contacts and locking devices, and the type of locking devices is set according to the requirements of the owner. The color of the cabinets is as required by the owner.
- 5.System of substation grounding type is TN-S.
- 6.Cabinet secondary circuit is set by the manufacturer combined with the measurement and control meter in the system diagram, and according to the manufacturer's product sample. This design is no longer drawn.
- 7.In low-voltage cabinet, frame breaker should have remote in/out position mechanical safety lock, molded case circuit breaker should have extension handle.
- 8.Low-voltage distribution cabinets of 630A and below use drawer type structure, of more than 630a use fixed separation type.
- 9.The two adjacent transformers are equipped with a low-voltage contact switch, and the mechanical interlock and electrical interlock are set between the contact switch and the main switch to ensure that only two of the three circuit breakers can be closed at the same time. The contact switch should be turned off when normal.
- 10.Set fire-proof partition on both sides of the fire circuit in the low-voltage cabinet.

02	2025.08.19	IFC-Connect 11W incoming cabinets
02	2025.04.03	IFC-Upgrade air conditioning etc.
02	2025.03.07	IFC
01	2024.12.07	FOR APPROVAL
建设单位 CLIENT		
Mega Motor Company (Private) Limited		
项目名称 PROJECT		
Mega Motor Company CXD Factory		
子项目名称 BUILDING		
焊装车间		
Body Welding Shop		
图纸名称 DRAWING TITLE		
1T4系统图		
1T4 schematics		
审定 AUTHORIZED		
设计负责人 DESIGN MANAGER		
审核 APPROVAL		
校对 CHECK		
设计负责人 DESIGNER (LEAD)		
设计 DESIGNER		
日期		
专业: DISCIPLINE	电气	阶段: STAGE
比例: SCALE	1 : 100	施工图
图号: DRAWING NO.	1409A5DQ1A-39G01	Revision: A

	PMU-53A	PMU-53A	PMU-53A	PMU-53A	PMU-53A	PMU-53A	PMU-53A
低压配电屏编号 LOW V. PANEL NUMBER	4ACC1	4ALL1	4AN4				4AN3
低压配电屏型号 LV PANEL TYPE	MNS	MNS	MNS				MNS
回路编号 NO. OF LOOP	--	--	4AN4-4	4AN4-3	4AN4-2	4AN4-1	4AN3-1
回路用途 USE OF CIRCUIT	动态无功补偿 Dynamic reactive power compensation	联络 connection	备用	备用	1APCD2	1APCD1	1T4-M1 (预留 for phase 2)
安装容量(kW) EQUIPMENT CAPACITY (kW)	TSC-A-405kVar (实际输出容量/Actual output capacity)	--	--	--	132.8	160	739
计算电流(A) CALCULATED CURRENT (A)	--	--	--	--	252.2	303.9	552.0
额定短路电流I _n (A) RATED CURRENT FOR RELEASE I _n (A)	2000	630	800	400	400	800	800
额定长延时电流I _t (A) RATED LONG-TERM CURRENT FOR RELEASE I _t (A)	1600	630	800	315	400	800	800
额定短时电流I _Δ (A) RATED SHORT-TERM CURRENT FOR RELEASE I _Δ (A)	5I _t (t=0.2S)	6I _t	6I _t	6I _t	OFF	OFF	OFF
额定瞬时电流I _r (A) RATED INSTANTANEOUS CURRENT FOR RELEASE I _r (A)	10I _t	10I _t	10I _t	10I _t	10I _t	10I _t	10I _t
额定接地故障电流I _g (A) RATED FAULT CURRENT FOR RELEASE I _g (A)	OFF	OFF	OFF	OFF	OFF	OFF	OFF
馈线规格 POWER CABLE	三相密集型母线槽2000A Three-phase dense busbar	--	--	YJV-4x185+1x95	2(YJV-4x95+1x50)	三相密集型母线槽800A Three-phase dense busbar	三相密集型母线槽800A Three-phase dense busbar
柜体尺寸(宽×高×深)mm CASE OF PANEL(W×H×D)mm	1000x1000x2200	1800	400	600	400	400	900
备注 REMARK	柜后设置机械通风(配套温度传感器) Mechanical ventilation (with temperature sensor) is set behind the cabinet.	上出线 going out from above	上出线 going out from above			上出线 going out from above	



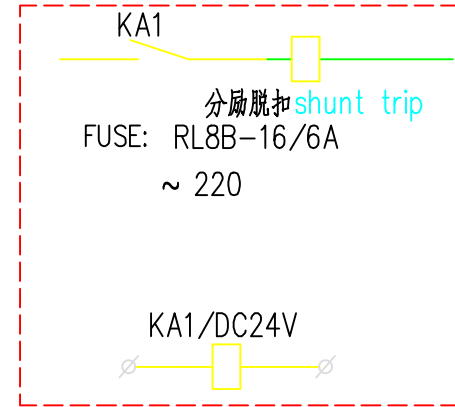
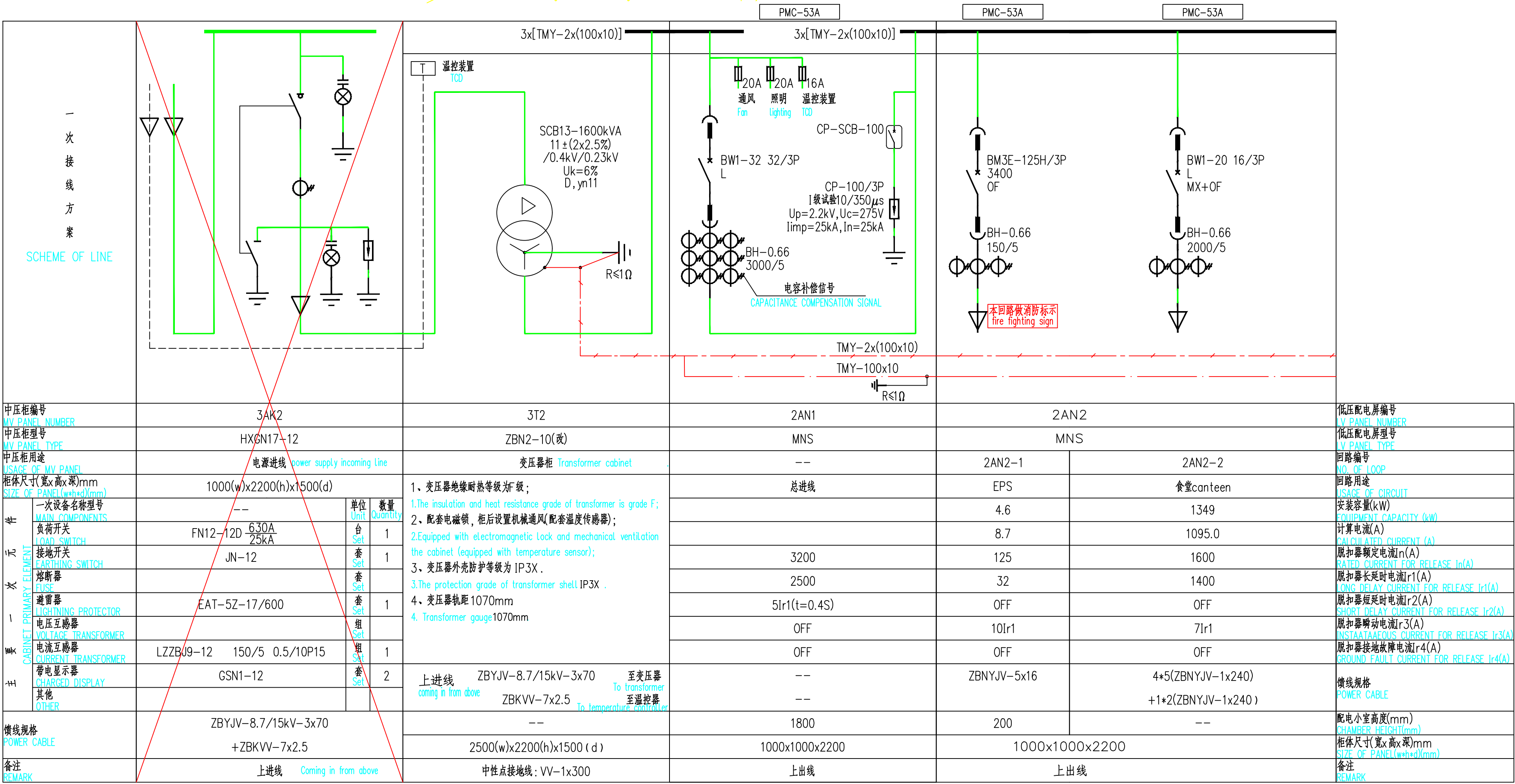
- 说明:
 1. 干式变压器采用[P33金防蚀户外壳]采用电缆进线, 出线接至出线柜, 接至出线柜位置与间距按《高压开关柜与母线柜配合》。
 2. 变压器高压侧(高压侧进线时), 变压器低压侧出线柜与高压柜(高压进线柜), 连接柜按《高压柜及出线柜配合》按ZBKVV1-10/2.5mm²。
 3. 变压器高压侧(高压侧进线), 电源来自低压柜柜前(如高压柜柜前), 高压柜按《高压柜、电容柜柜前及出线柜配合》。
 4. 变压器接自低压柜柜前(如高压柜柜前)。
 5. 变压器柜门边设有带电危险警告。
 6. 柜内上下层为生产厂家标准柜体本柜按国家标准设计, 按照生产厂家标准本柜定制, 本设计不作为图出。
 7. 低压柜(柜内)内部按厂家标配制造, 输出位置按厂家全接, 警告标志柜前柜后柜上。
 8. 柜内柜前柜后30A以下及100A以下按柜体定制大7630A柜内固定分断式。
 9. 柜内柜前柜后回路带短路保护装置。
 10. 柜前柜后柜上温度测量, 柜内柜后温度测量点, 且能在柜上运行查看。

精装修改，取消此回路
This circuit is canceled per the fit-out revision.

1. **10-type** transformer (with IP5X metal protective shell) use cable lead line, horizontal busbar side outlet, horizontal busbar outlet position and spacing should be matched with the low-voltage cabinet bus.
2. Transformer has a temperature controller (forced air cooling), whose power is from the special fuse added before the main switch of the low-voltage side. The temperature alarm and trip signal lead to high-voltage switchgear using HW-7x2.5mm².
3. Set lighting in the transformer cabinet, whose power is from the special fuse added before the main switch of the low-voltage side. Set mechanical ventilation on the back of the transformer cabinet and capacitor compensation cabinet, whose power is from the special fuse added before the main switch of the low-voltage side.
4. The transformer cabinet door should have a live lock function, and the door should be locked when it is live; All circuit breakers are equipped with auxiliary contacts and locking devices, and the type of locking devices is set according to the requirements of the owner. The color of the cabinets is as required by the owner.
5. System of substitution grounding type is TN-S.
6. Cabinet secondary circuit is set by the manufacturer combined with the measurement and control meter in the system diagram, and according to the manufacturer's product sample. This design is no longer demand.
7. In low-voltage circuit, frame breaker should have remote in/out position mechanical safety lock, molded case circuit breaker should have extension handle.
8. Low-voltage distribution cabinets of 630A and below use drawer type structure, of more than 630A use the separation type.
9. Set fire circuit protection on both sides of the fire circuit in the low-voltage cabinet.
10. The capacitor should be equipped with a temperature detection device. The temperature status should be displayed on the cabinet door, and it should be able to send out warnings on mobile phones.

1409A5DQ3A-28G03

2025.08.19: 根据甲方要求取消11kV进线柜
Cancel 11kV incoming cabinets according to the demand of the first party



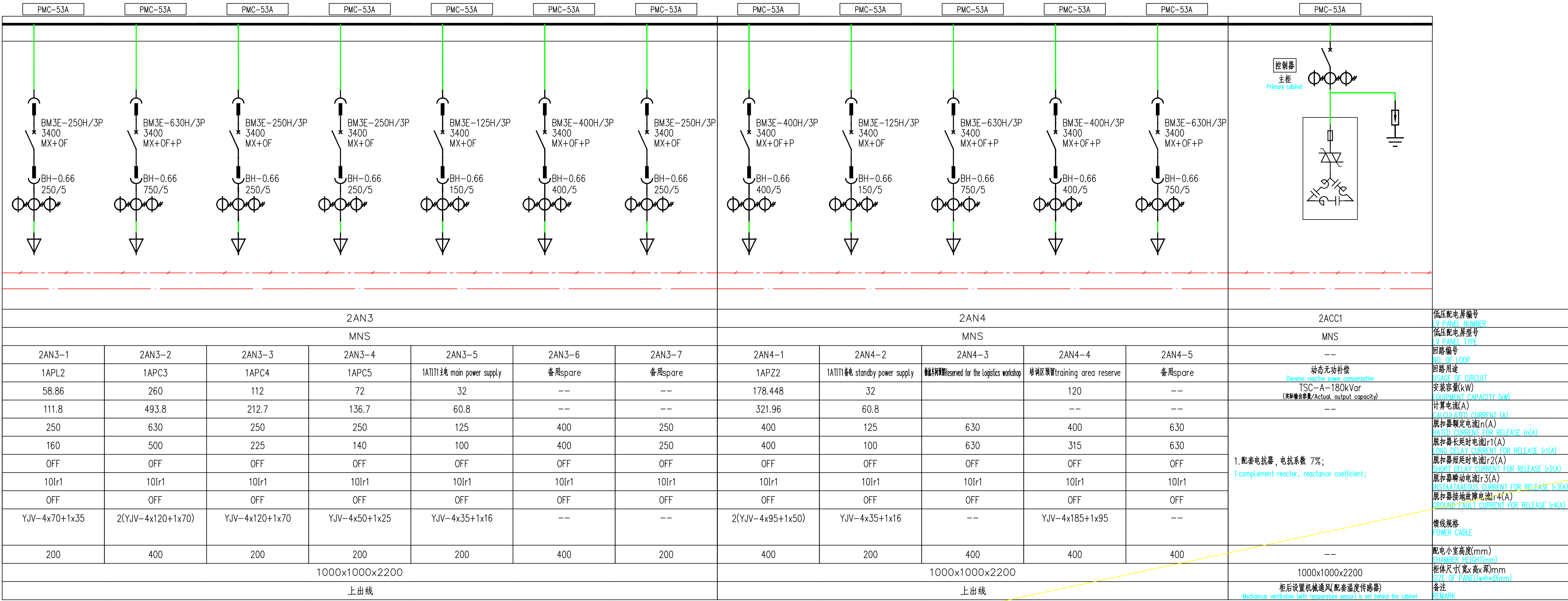
Secondary diagram of non-fire power supply shunting off
非消防电源切除二次图

适用于低压柜所有带分励脱扣的回路
Suitable for all circuits with shunt trip in low voltage cabinet

- 说明:
- 干式变压器(带IP3X金属防护外壳)采用电缆进线, 母线槽引出线, 母线出线位置与间距应与低压屏母线相配合。
 - 变压器带温控器(强迫风冷), 电源接自低压柜总开关前(加装专用熔断器), 温控器温度报警及跳闸信号采用ZBKVV-7x2.5mm²。
 - 变压器柜内设置照明, 电源接自低压柜总开关前(加装专用熔断器), 电源接自低压柜总开关前(加装专用熔断器)。
 - 变压器柜门应具有带电闭锁功能, 变压器带电时应闭锁; 图中所有断路器均配辅助触点和锁定装置, 额定装置失压脱压主要设置, 配电柜柜体颜色按业主要求。
 - 变电所接地型式采用TN-S制。
 - 柜内二次线路由生产厂家综合本系统图中的测控表计, 按照生产厂家产品样本制定, 本设计不再出图。
 - 低压配电屏内框架开关均配进线/出线位置机械安全锁, 警示断路器旁设手柄。
 - 低压配电柜630A及以下采用抽屉式结构, 大于630A采用固定分隔式。
 - 高低压柜进线柜两侧设置防火门。
 - 电容器要设置温度检测, 在柜门上设置温度状态, 且能在手机上进行报警。

Note:

- Dry-type transformer (with IP3X metal protective shell) use cable lead line, horizontal busbar side outlet, horizontal busbar outlet position and spacing should be matched with the low-voltage cabinet bus.
- Transformer has a temperature controller (forced air cooling), whose power is from the special fuse added before the main switch of the low-voltage side. The thermostat temperature alarm and trip signal lead to high-voltage switchgear using kVV-7x2.5mm².
- Set lighting in the transformer cabinet, whose power is from the special fuse added before the main switch of the low-voltage side. Set mechanical ventilation on the back of the transformer cabinet and capacitor compensation cabinet, whose power is from the special fuse added before the main switch of the low-voltage side.
- The transformer cabinet door should have a live lock function, and the door should be locked when it is live; All circuit breakers are equipped with auxiliary contacts and locking devices, and the type of locking devices is set according to the requirements of the owner. The color of the cabinets is as required by the owner.
- System of substation grounding type is TN-S.
- Cabinet secondary circuit is set by the manufacturer combined with the measurement and control meter in the system diagram, and according to the manufacturer's product sample. This design is no longer drawn.
- In low-voltage cabinet, frame breaker should have remote in/out position mechanical safety lock, molded case circuit breaker should have extension handle.
- Low voltage distribution cabinets of 630A and below use drawer type structure; 630A and below use fixed separation type.
- Set fire-proof partition on both sides of the fire circuit in the low voltage cabinet.
- The capacitor should be equipped with a temperature detection device. The temperature status should be displayed on the cabinet door, and it should be able to send out warnings on mobile phones.



AE
中国汽车工业工程有限公司
AUTOMOTIVE ENGINEERING CORPORATION

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2. 本图压和相关专业图样配合使用。

说明或图例 DESCRIPTIONS OR ILLUSTRATIONS

图例 Legend
符号 meaning
SD 报警触头 alarm contact
MX 分励脱扣器 shunt trip
OF 辅助触头 auxiliary contact
MN 欠压脱扣器 undervoltage release
P 电动机操作机构 electric operating mechanism
Z 手动操作机构 manual operating mechanism

06	2026.01.13	IFC- IT Change
07	2026.11.13	IFC- Architecture Change
08	2026.10.30	IFC-Decoration Change
09	2026.10.21	IFC-Decoration Change
04	2026.08.20	IFC-Cancel 11kV incoming cabinets
03	2026.04.03	IFC-Update air conditioning,etc.
02	2025.03.07	IFC
01	2024.12.13	FOR APPROVAL

建设单位 CLIENT

Mega Motor Company (Private) Limited

项目名称 PROJECT

Mega Motor Company CKD Factory

子项名称 BUILDING

General Assembly Shop

图样名称 DRAWING TITLE

3T2 变压器系统图
3T2 schematics

审 定 AUTHORIZED 设计负责人 DESIGN MANAGER		
审 核 APPROVAL		
校 对 CHECK		
设计负责人 DISCIPLINE LEAD		
设 计 DESIGNER		
日 期 DATE		
专 业 DISCIPLINE	电气	阶段: 施工图 STAGE
比 例: SCALE	1:100	版 次: REVISION
图 号: DRAWING NO.	1409A5DQ3A-29G03	